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$$\begin{aligned} & \lim_{x \rightarrow 0} \frac{x \ln(1+x)}{e^x - x - 1} \\ &= \frac{0}{0} \stackrel{H}{=} \lim_{x \rightarrow 0} \frac{\ln(1+x) + \frac{x}{1+x}}{e^x - 1} \\ &= \frac{0}{0} \stackrel{H}{=} \lim_{x \rightarrow 0} \frac{\frac{1}{1+x} + \frac{1+x-x}{(1+x)^2}}{e^x} \\ &= 2 \end{aligned}$$

References